

SIMATS ENGINEERING

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Course Code: CSA1707

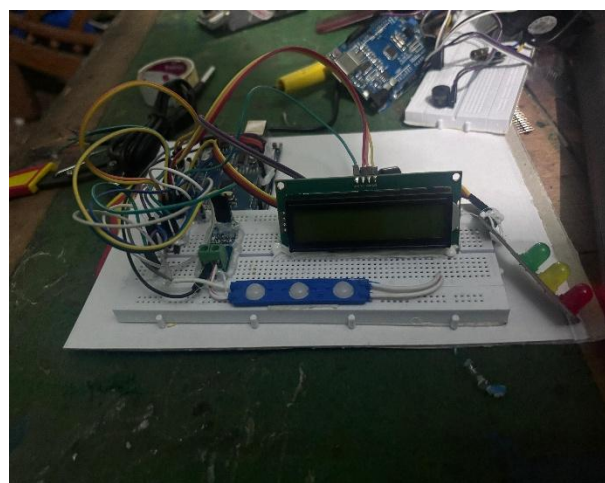
Slot:B

Course Name: ARTIFICIAL INTELLIGENCE

Course Faculty: DR.R.VEERAMANI

Project Title: AI-Powered Electrical Panel Fault Prediction and Preventive Maintenance System

Module Photographs:



Project Description:

The **AI-Powered Electrical Panel Fault Prediction and Preventive Maintenance System** is an intelligent system designed to monitor electrical panels, identify abnormal operating conditions, predict possible faults, and support preventive maintenance before serious failures occur. The system collects important electrical parameters such as voltage, current, temperature, power consumption, and other sensor readings from electrical panels and analyzes the collected data using **Artificial Intelligence and Machine Learning techniques**. By studying historical and real-time patterns, the system can detect unusual behavior and predict potential faults such as overheating, overload, voltage fluctuations, short circuits, and other electrical abnormalities. When a possible fault is identified, the system generates an alert and provides information about the fault condition, allowing maintenance personnel to take corrective action at an early stage. The system also helps in scheduling preventive maintenance based on the predicted condition of the electrical panel rather than relying only on fixed maintenance intervals.

STUDENT SIGNATURE

V.LASYA

GUIDE SIGNATURE

DR . R . VEERAMANI